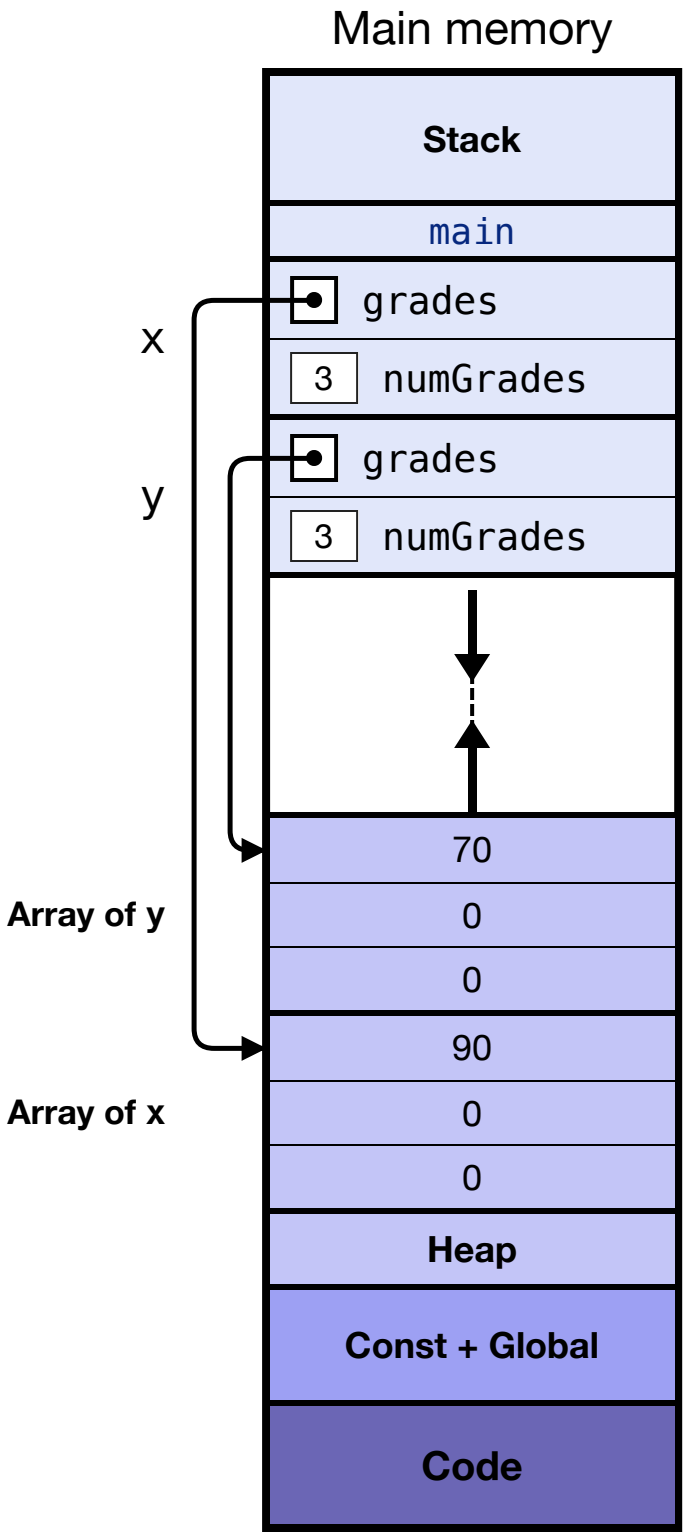
Our copy constructor allocates a new array for y and copies the grades of x into it




```
class Student {
  private:
    int* grades;
    int numGrades;
  public:
    Student(int n) {
      numGrades = n;
      grades = new int[numGrades];
      // ... set all grades to 0
    }
    Student(const Student& other) {
      numGrades = other.numGrades;
      grades = new int[numGrades];
      // ... copy each grade
    }
    ~Student() {
      delete[] grades;
    }
};

int main(void) {
  Student x(3);
  x.setGrade(0, 90);
  Student y = x;
  y.setGrade(0, 70);
  return 0;
}
```